

NTA JEE Mains Jan 2026

Test Date	28/01/2026
Test Time	3:00 PM - 6:00 PM
Subject	B. Tech

Section : Mathematics Section A

Q.1

Let the arithmetic mean of $\frac{1}{a}$ and $\frac{1}{b}$ be $\frac{5}{16}$, $a > 2$. If α is such that $a, 4, \alpha, b$ are in A.P., then the equation $\alpha x^2 - ax + 2(\alpha - 2b) = 0$ has :

Options

1. both roots in the interval $(-2, 0)$
2. one root in $(0, 2)$ and another in $(-4, -2)$
3. complex roots of magnitude less than 2
4. one root in $(1, 4)$ and another in $(-2, 0)$

Question Type : **MCQ**

Question ID : **8606541654**

Option 1 ID : **8606545624**

Option 2 ID : **8606545626**

Option 3 ID : **8606545623**

Option 4 ID : **8606545625**

Q.2

Given below are two statements :

Statement I : The function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{x}{1 + |x|}$ is one-one.

Statement II : The function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{x^2 + 4x - 30}{x^2 - 8x + 18}$ is many-one.

In the light of the above statements, choose the **correct answer** from the options given below :

Options

1. **Statement I is false but Statement II is true**
2. **Both Statement I and Statement II are true**
3. **Both Statement I and Statement II are false**
4. **Statement I is true but Statement II is false**

Question Type : **MCQ**

Question ID : **8606541652**

Option 1 ID : **8606545618**

Option 2 ID : **8606545615**

Option 3 ID : **8606545616**

Option 4 ID : **8606545617**

Q.3

Let

$$A = \{z \in \mathbb{C} : |z - 2| \leq 4\} \text{ and}$$

$$B = \{z \in \mathbb{C} : |z - 2| + |z + 2| = 5\}.$$

Then the max $\{|z_1 - z_2| : z_1 \in A \text{ and } z_2 \in B\}$ is :

Options

1. 9
2. $\frac{15}{2}$
3. $\frac{17}{2}$
4. 8

Question Type : MCQ

Question ID : 8606541653

Option 1 ID : 8606545622

Option 2 ID : 8606545619

Option 3 ID : 8606545621

Option 4 ID : 8606545620

Q.4

Given below are two statements :

Statement I : $25^{13} + 20^{13} + 8^{13} + 3^{13}$ is divisible by 7.

Statement II : The integral part of $(7 + 4\sqrt{3})^{25}$ is an odd number.

In the light of the above statements, choose the **correct answer** from the options given below :

Options

1. **Statement I** is true but **Statement II** is false
2. Both **Statement I** and **Statement II** are true
3. **Statement I** is false but **Statement II** is true
4. Both **Statement I** and **Statement II** are false

Question Type : MCQ

Question ID : 8606541657

Option 1 ID : 8606545637

Option 2 ID : 8606545635

Option 3 ID : 8606545638

Option 4 ID : 8606545636

Q.5 An ellipse has its center at $(1, -2)$, one focus at $(3, -2)$ and one vertex at $(5, -2)$. Then the length of its latus rectum is :

Options

1. $4\sqrt{3}$
2. $6\sqrt{3}$
3. 6
4. $\frac{16}{\sqrt{3}}$

Question Type : MCQ

Question ID : 8606541659

Option 1 ID : 8606545645

Option 2 ID : 8606545643

Option 3 ID : 8606545646

Option 4 ID : 8606545644

Q.6 Let A be the focus of the parabola $y^2 = 8x$. Let the line $y = mx + c$ intersect the parabola at two distinct points B and C. If the centroid of the triangle ABC is $\left(\frac{7}{3}, \frac{4}{3}\right)$, then $(BC)^2$ is equal to :

Options

1. 41
2. 80
3. 32
4. 89

Question Type : MCQ

Question ID : 8606541663

Option 1 ID : 8606545660

Option 2 ID : 8606545661

Option 3 ID : 8606545659

Option 4 ID : 8606545662

Q.7 Considering the principal values of inverse trigonometric functions, the value of the expression

$$\tan\left(2\sin^{-1}\left(\frac{2}{\sqrt{13}}\right) - 2\cos^{-1}\left(\frac{3}{\sqrt{10}}\right)\right) \text{ is equal to :}$$

Options

1. $-\frac{33}{56}$
2. $\frac{33}{56}$
3. $\frac{16}{63}$
4. $-\frac{16}{63}$

Question Type : **MCQ**

Question ID : **8606541661**

Option 1 ID : **8606545654**

Option 2 ID : **8606545652**

Option 3 ID : **8606545651**

Option 4 ID : **8606545653**

Q.8 The sum of all the elements in the range of $f(x) = \text{Sgn}(\sin x) + \text{Sgn}(\cos x) + \text{Sgn}(\tan x) + \text{Sgn}(\cot x)$,

$$x \neq \frac{n\pi}{2}, n \in \mathbb{Z}, \text{ where } \text{Sgn}(t) = \begin{cases} 1, & \text{if } t > 0 \\ -1, & \text{if } t < 0 \end{cases}, \text{ is :}$$

Options

1. **0**
2. **-2**
3. **2**
4. **4**

Question Type : **MCQ**

Question ID : **8606541651**

Option 1 ID : **8606545614**

Option 2 ID : **8606545613**

Option 3 ID : **8606545611**

Option 4 ID : **8606545612**

Q.9

Let $y = y(x)$ be the solution of the differential equation $x \frac{dy}{dx} - y = x^2 \cot x$, $x \in (0, \pi)$. If $y\left(\frac{\pi}{2}\right) = \frac{\pi}{2}$, then

$6y\left(\frac{\pi}{6}\right) - 8y\left(\frac{\pi}{4}\right)$ is equal to :

Options

1. -3π
2. π
3. 3π
4. $-\pi$

Question Type : **MCQ**

Question ID : **8606541669**

Option 1 ID : **8606545685**

Option 2 ID : **8606545684**

Option 3 ID : **8606545686**

Option 4 ID : **8606545683**

Q.10

Let $[\cdot]$ denote the greatest integer function. Then $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{12(3+[x])}{3+[\sin x]+[\cos x]} \right) dx$ is equal to :

Options

1. $13\pi + 1$
2. $11\pi + 2$
3. $15\pi + 4$
4. $12\pi + 5$

Question Type : **MCQ**

Question ID : **8606541667**

Option 1 ID : **8606545676**

Option 2 ID : **8606545677**

Option 3 ID : **8606545675**

Option 4 ID : **8606545678**

Q.11

Let $f(x) = \int \frac{dx}{x^{\left(\frac{2}{3}\right)} + 2x^{\left(\frac{1}{2}\right)}}$ be such that $f(0) = -26 + 24 \log_e(2)$. If $f(1) = a + b \log_e(3)$, where $a, b \in \mathbb{Z}$,

then $a + b$ is equal to :

Options

1. -5
2. -11
3. -26
4. -18

Question Type : **MCQ**

Question ID : **8606541668**

Option 1 ID : **8606545682**

Option 2 ID : **8606545681**

Option 3 ID : **8606545679**

Option 4 ID : **8606545680**

Q.12

Let Q(a, b, c) be the image of the point P(3, 2, 1) in the line $\frac{x-1}{1} = \frac{y}{2} = \frac{z-1}{1}$. Then the distance of

Q from the line $\frac{x-9}{3} = \frac{y-9}{2} = \frac{z-5}{-2}$ is

Options

1. 5
2. 8
3. 6
4. 7

Question Type : MCQ

Question ID : 8606541665

Option 1 ID : 8606545668

Option 2 ID : 8606545667

Option 3 ID : 8606545669

Option 4 ID : 8606545670

Q.13

The sum of the coefficients of x^{499} and x^{500} in $(1+x)^{1000} + x(1+x)^{999} + x^2(1+x)^{998} + \dots + x^{1000}$ is :

Options

1. $^{1000}C_{501}$
2. $^{1002}C_{500}$
3. $^{1002}C_{501}$
4. $^{1001}C_{501}$

Question Type : MCQ

Question ID : 8606541656

Option 1 ID : 8606545633

Option 2 ID : 8606545632

Option 3 ID : 8606545631

Option 4 ID : 8606545634

Q.14

Let the ellipse E : $\frac{x^2}{144} + \frac{y^2}{169} = 1$ and the hyperbola H : $\frac{x^2}{16} - \frac{y^2}{\lambda^2} = -1$ have the same foci. If e and L

respectively denote the eccentricity and the length of the latus rectum of H, then the value of $24(e+L)$ is :

Options

1. 67
2. 296
3. 126
4. 148

Question Type : MCQ

Question ID : 8606541662

Option 1 ID : 8606545655

Option 2 ID : 8606545658

Option 3 ID : 8606545656

Option 4 ID : 8606545657

Q.15 Let the circle $x^2 + y^2 = 4$ intersect x -axis at the points $A(a, 0)$, $a > 0$ and $B(b, 0)$. Let $P(2 \cos \alpha, 2 \sin \alpha)$, $0 < \alpha < \frac{\pi}{2}$ and $Q(2 \cos \beta, 2 \sin \beta)$ be two points such that $(\alpha - \beta) = \frac{\pi}{2}$. Then the point of intersection of AQ and BP lies on :

Options

1. $x^2 + y^2 - 4x - 4y = 0$
2. $x^2 + y^2 - 4y - 4 = 0$
3. $x^2 + y^2 - 4x - 4 = 0$
4. $x^2 + y^2 - 4x - 4y - 4 = 0$

Question Type : **MCQ**

Question ID : **8606541660**

Option 1 ID : **8606545650**

Option 2 ID : **8606545648**

Option 3 ID : **8606545647**

Option 4 ID : **8606545649**

Q.16 The probability distribution of a random variable X is given below :

X	$4k$	$\frac{30}{7}k$	$\frac{32}{7}k$	$\frac{34}{7}k$	$\frac{36}{7}k$	$\frac{38}{7}k$	$\frac{40}{7}k$	$6k$
$P(X)$	$\frac{2}{15}$	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{1}{5}$	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{1}{5}$	$\frac{1}{15}$

If $E(X) = \frac{263}{15}$, then $P(X < 20)$ is equal to :

Options

1. $\frac{3}{5}$
2. $\frac{8}{15}$
3. $\frac{11}{15}$
4. $\frac{14}{15}$

Question Type : **MCQ**

Question ID : **8606541658**

Option 1 ID : **8606545640**

Option 2 ID : **8606545639**

Option 3 ID : **8606545641**

Option 4 ID : **8606545642**

Q.17

$$\frac{6}{3^{26}} + \frac{10 \cdot 1}{3^{25}} + \frac{10 \cdot 2}{3^{24}} + \frac{10 \cdot 2^2}{3^{23}} + \dots + \frac{10 \cdot 2^{24}}{3}$$
 is equal to :

Options

1. 2^{26}
2. 2^{25}
3. 3^{25}
4. 3^{26}

Question Type : MCQ

Question ID : 8606541655

Option 1 ID : 8606545630

Option 2 ID : 8606545629

Option 3 ID : 8606545627

Option 4 ID : 8606545628

Q.18

Let $P_1 : y = 4x^2$ and $P_2 : y = x^2 + 27$ be two parabolas. If the area of the bounded region enclosed between P_1 and P_2 is six times the area of the bounded region enclosed between the line $y = \alpha x$, $\alpha > 0$ and P_1 , then α is equal to :

Options

1. 12
2. 15
3. 6
4. 8

Question Type : MCQ

Question ID : 8606541670

Option 1 ID : 8606545689

Option 2 ID : 8606545690

Option 3 ID : 8606545687

Option 4 ID : 8606545688

Q.19 Let P be a point in the plane of the vectors $\vec{AB} = 3\hat{i} + \hat{j} - \hat{k}$ and $\vec{AC} = \hat{i} - \hat{j} + 3\hat{k}$ such that P is equidistant

from the lines AB and AC. If $\left| \vec{AP} \right| = \frac{\sqrt{5}}{2}$, then the area of the triangle ABP is :

Options

1. $\frac{\sqrt{30}}{4}$
2. $\frac{\sqrt{26}}{4}$
3. 2
4. $\frac{3}{2}$

Question Type : MCQ

Question ID : 8606541664

Option 1 ID : 8606545665

Option 2 ID : 8606545664

Option 3 ID : 8606545666

Option 4 ID : 8606545663

Q.20

Let $f(x) = \lim_{\theta \rightarrow 0} \left(\frac{\cos \pi x - x^{\left(\frac{2}{\theta}\right)} \sin(x-1)}{1 + x^{\left(\frac{2}{\theta}\right)} (x-1)} \right), x \in \mathbf{R}$. Consider the following two statements :

(I) $f(x)$ is discontinuous at $x = 1$.

(II) $f(x)$ is continuous at $x = -1$.

Then,

Options

1. Neither (I) nor (II) is True
2. Both (I) and (II) are True
3. Only (I) is True
4. Only (II) is True

Question Type : MCQ

Question ID : 8606541666

Option 1 ID : 8606545674

Option 2 ID : 8606545673

Option 3 ID : 8606545671

Option 4 ID : 8606545672

Section : Mathematics Section B

Q.21

If the distance of the point $P(43, \alpha, \beta)$, $\beta < 0$, from the line $\vec{r} = 4\hat{i} - \hat{k} + \mu(2\hat{i} + 3\hat{k})$, $\mu \in \mathbf{R}$ along a line with direction ratios 3, -1, 0 is $13\sqrt{10}$, then $\alpha^2 + \beta^2$ is equal to _____

Question Type : SA

Question ID : 8606541674

Q.22 Three persons enter in a lift at the ground floor. The lift will go upto 10th floor. The number of ways, in which the three persons can exit the lift at three different floors, if the lift does not stop at first, second and third floors, is equal to _____.

Question Type : SA

Question ID : 8606541673

Q.23 If $\sum_{r=1}^{25} \left(\frac{r}{r^4 + r^2 + 1} \right) = \frac{p}{q}$, where p and q are positive integers such that $\gcd(p, q) = 1$, then p + q is equal to _____.

Question Type : SA

Question ID : 8606541672

Q.24 Let f be a differentiable function satisfying $f(x) = 1 - 2x + \int_0^x e^{(x-t)} f(t) dt, x \in \mathbb{R}$ and let

$g(x) = \int_0^x (f(t) + 2)^{15} (t - 4)^6 (t + 12)^{17} dt, x \in \mathbb{R}$. If p and q are respectively the points of local minima and local maxima of g, then the value of |p + q| is equal to _____.

Question Type : SA

Question ID : 8606541675

Q.25 Let $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$ and B be two matrices such that $A^{100} = 100B + I$. Then the sum of all the elements of B^{100} is _____

Question Type : SA

Question ID : 8606541671

Section : Physics Section A

Q.26 A plane electromagnetic wave is moving in free space with velocity $c = 3 \times 10^8$ m/s and its electric field is given as $\vec{E} = 54 \sin(kz - \omega t) \hat{j}$ V/m, where $\hat{j}$ is the unit vector along y-axis. The magnetic field vector $\vec{B}$ of the wave is :

Options

1. $1.4 \times 10^{-7} \sin(kz - \omega t) \hat{k}$ T
2. $+1.8 \times 10^{-7} \sin(kz - \omega t) \hat{i}$ T
3. $-1.8 \times 10^{-7} \sin(kz - \omega t) \hat{i}$ T
4. $1.4 \times 10^{-7} \sin(kz - \omega t) \hat{i}$ T

Question Type : MCQ

Question ID : 8606541690

Option 1 ID : 8606545753

Option 2 ID : 8606545754

Option 3 ID : 8606545752

Option 4 ID : 8606545755

Q.27 When the position vector $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ changes sign as $-\vec{r}$, which one of the following vector will not flip under sign change?

Options

1. Velocity
2. Angular momentum
3. Linear momentum
4. Acceleration

Question Type : MCQ

Question ID : 8606541681

Option 1 ID : 8606545716

Option 2 ID : 8606545718

Option 3 ID : 8606545719

Option 4 ID : 8606545717

Q.28 A nucleus has mass number α and radius R_α . Another nucleus has mass number β and radius R_β . If $\beta = 8\alpha$ then R_α / R_β is :

Options

1. 1
2. 8
3. 0.5
4. 2

Question Type : MCQ

Question ID : 8606541694

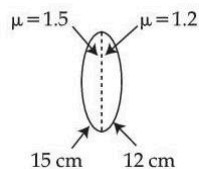
Option 1 ID : 8606545769

Option 2 ID : 8606545770

Option 3 ID : 8606545768

Option 4 ID : 8606545771

- Q.29** A biconvex lens is formed by using two thin planoconvex lenses, as shown in the figure. The refractive index and radius of curved surfaces are also mentioned in figure. When an object is placed on the left side of lens at a distance of 30 cm from the biconvex lens, the magnification of the image will be :



Options

1. -2
2. $+2$
3. $+2.5$
4. -2.5

Question Type : **MCQ**

Question ID : **8606541692**

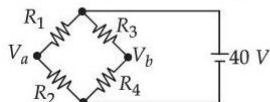
Option 1 ID : **8606545761**

Option 2 ID : **8606545760**

Option 3 ID : **8606545763**

Option 4 ID : **8606545762**

- Q.30** A Wheatstone bridge is initially at room temperature and all arms of the bridge have same value of resistances ($R_1 = R_2 = R_3 = R_4$). When R_3 resistance is heated to some temperature, its resistance value has gone up by 10%. The potential difference ($V_a - V_b$) (after R_3 is heated) is _____ V.



Options

1. 0
2. 0.95
3. 1.05
4. 2

Question Type : **MCQ**

Question ID : **8606541687**

Option 1 ID : **8606545741**

Option 2 ID : **8606545743**

Option 3 ID : **8606545740**

Option 4 ID : **8606545742**

Q.31 Identify the correct statements :

- A. Electrostatic field lines form closed loops .
- B. The electric field lines point radially outward when charge is greater than zero.
- C. The Gauss - Law is valid only for inverse - square force.
- D. The workdone in moving a charged particle in a static electric field around a closed path is zero.
- E. The motion of a particle under Coulomb's force must take place in a plane.

Choose the **correct** answer from the options given below :

Options

- 1. B, C, D, E Only
- 2. A, B, D, E Only
- 3. A, C, E Only
- 4. A, B, C, D Only

Question Type : **MCQ**

Question ID : **8606541685**

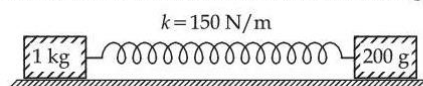
Option 1 ID : **8606545733**

Option 2 ID : **8606545735**

Option 3 ID : **8606545734**

Option 4 ID : **8606545732**

Q.32 As shown in the figure, a spring is kept in a stretched position with some extension by holding the masses 1 kg and 0.2 kg with a separation more than spring natural length and are released. Assuming the horizontal surface to be frictionless, the angular frequency (in SI unit) of the system is :



Options

- 1. 20
- 2. 30
- 3. 5
- 4. 27

Question Type : **MCQ**

Question ID : **8606541678**

Option 1 ID : **8606545707**

Option 2 ID : **8606545705**

Option 3 ID : **8606545704**

Option 4 ID : **8606545706**

Q.33 Number of photons of equal energy emitted per second by a 6 mW laser source operating at 663 nm is _____. (Given : $h = 6.63 \times 10^{-34}$ J.s and $c = 3 \times 10^8$ m/s)

Options

1. 5×10^{15}
2. 5×10^{16}
3. 2×10^{16}
4. 10×10^{15}

Question Type : **MCQ**

Question ID : **8606541693**

Option 1 ID : **8606545764**

Option 2 ID : **8606545767**

Option 3 ID : **8606545766**

Option 4 ID : **8606545765**

Q.34 A particle starts moving from time $t=0$ and its coordinate is given as $x(t) = 4t^3 - 3t$

- A. The particle returns to its original position (origin) 0.866 units later
- B. The particle is 1 unit away from origin at its turning point
- C. Acceleration of the particle is non-negative
- D. The particle is 0.5 units away from origin at its turning point
- E. Particle never turns back as acceleration is non-negative

Choose the **correct** answer from the options given below :

Options

1. C, E Only
2. A, B, C Only
3. A, C Only
4. A, C, D Only

Question Type : **MCQ**

Question ID : **8606541680**

Option 1 ID : **8606545715**

Option 2 ID : **8606545714**

Option 3 ID : **8606545712**

Option 4 ID : **8606545713**

Q.35 Identify the correct statements :

- A. Effective capacitance of a series combination of capacitors is always smaller than the smallest capacitance of the capacitor in the combination.
- B. When a dielectric medium is placed between the charged plates of a capacitor, displacement of charges cannot occur due to insulation property of dielectric.
- C. Increasing of area of capacitor plate or decreasing of thickness of dielectric is an alternate method to increase the capacitance.
- D. For a point charge, concentric spherical shells centered at the location of the charge are equipotential surfaces.

Choose the **correct** answer from the options given below :

Options

- 1. C and D Only
- 2. A, C and D Only
- 3. B and D Only
- 4. A, B and C Only

Question Type : **MCQ**

Question ID : **8606541686**

Option 1 ID : **8606545738**

Option 2 ID : **8606545739**

Option 3 ID : **8606545737**

Option 4 ID : **8606545736**

Q.36

The time period of a simple harmonic oscillator is $T = 2\pi\sqrt{\frac{k}{m}}$. Measured value of mass (m) of the object is 10 g with an accuracy of 10 mg and time for 50 oscillations of the spring is found to be 60 s using a watch of 2 s resolution. Percentage error in determination of spring constant (k) is _____%.

Options

- 1. 6.76
- 2. 3.43
- 3. 7.60
- 4. 3.35

Question Type : **MCQ**

Question ID : **8606541677**

Option 1 ID : **8606545700**

Option 2 ID : **8606545703**

Option 3 ID : **8606545702**

Option 4 ID : **8606545701**

Q.37 In an experiment, a set of reading are obtained as follows - 1.24 mm, 1.25 mm, 1.23 mm, 1.21 mm. The expected least count of the instrument used in recording these readings is _____ mm.

Options

1. 0.001
2. 0.1
3. 0.01
4. 0.05

Question Type : MCQ

Question ID : 8606541676

Option 1 ID : 8606545697

Option 2 ID : 8606545696

Option 3 ID : 8606545699

Option 4 ID : 8606545698

Q.38 A long cylindrical conductor with large cross section carries an electric current distributed uniformly over its cross-section. Magnetic field due to this current is :

- A. maximum at either ends of the conductor and minimum at the midpoint
- B. maximum at the axis of the conductor
- C. minimum at the surface of the conductor
- D. minimum at the axis of the conductor
- E. same at all points in the cross-section of the conductor

Choose the **correct** answer from the options given below :

Options

1. B, C Only
2. D Only
3. E Only
4. A, D Only

Question Type : MCQ

Question ID : 8606541688

Option 1 ID : 8606545746

Option 2 ID : 8606545745

Option 3 ID : 8606545747

Option 4 ID : 8606545744

Q.39

Match List - I with List - II.

List - I

- A. Coefficient of viscosity
 B. Surface tension
 C. Pressure
 D. Surface energy

List - II

- I. $[ML^{-1}T^{-2}]$
 II. $[ML^2T^{-2}]$
 III. $[ML^0T^{-2}]$
 IV. $[ML^{-1}T^{-1}]$

Choose the **correct** answer from the options given below :

Options

1. A-IV, B-III, C-I, D-II
2. A-IV, B-I, C-II, D-III
3. A-I, B-II, C-IV, D-III
4. A-I, B-III, C-II, D-IV

Question Type : **MCQ**Question ID : **8606541682**Option 1 ID : **8606545720**Option 2 ID : **8606545723**Option 3 ID : **8606545721**Option 4 ID : **8606545722**

Q.40

The mean free path of a molecule of diameter 5×10^{-10} m at the temperature 41°C and pressure 1.38×10^5 Pa, is given as _____ m. (Given $k_B = 1.38 \times 10^{-23}$ J/K).

Options

1. $10\sqrt{2} \times 10^{-8}$
2. $2\sqrt{2} \times 10^{-8}$
3. $2\sqrt{2} \times 10^{-10}$
4. 2×10^{-8}

Question Type : **MCQ**Question ID : **8606541684**Option 1 ID : **8606545728**Option 2 ID : **8606545729**Option 3 ID : **8606545730**Option 4 ID : **8606545731**

Q.41 The speed of a longitudinal wave in a metallic bar is 400 m/s. If the density and Young's modulus of the bar material are increased by 0.5% and 1%, respectively then the speed of the wave is changed approximately to _____ m/s.

Options

1. 401
2. 402
3. 399
4. 398

Question Type : **MCQ**

Question ID : **8606541683**

Option 1 ID : **8606545725**

Option 2 ID : **8606545726**

Option 3 ID : **8606545727**

Option 4 ID : **8606545724**

Q.42 Which one of the following is **not** a measurable quantity ?

Options

1. Voltage
2. Voltage difference
3. Displacement current
4. Resistance

Question Type : **MCQ**

Question ID : **8606541689**

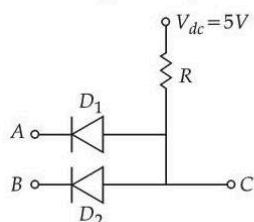
Option 1 ID : **8606545749**

Option 2 ID : **8606545750**

Option 3 ID : **8606545748**

Option 4 ID : **8606545751**

- Q.43** Two p-n junction diodes D_1 and D_2 are connected as shown in figure. A and B are input signals and C is the output. The given circuit will function as a _____.



- Options
1. OR Gate
 2. NOR Gate
 3. AND Gate
 4. NAND Gate

Question Type : MCQ

Question ID : 8606541695

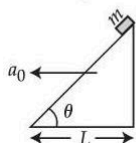
Option 1 ID : 8606545772

Option 2 ID : 8606545775

Option 3 ID : 8606545774

Option 4 ID : 8606545773

- Q.44** A small block of mass m slides down from the top of a frictionless inclined surface, while the inclined plane is moving towards left with constant acceleration a_0 . The angle between the inclined plane and ground is θ and its base length is L . Assuming that initially the small block is at the top of the inclined plane, the time it takes to reach the lowest point of the inclined plane is _____.



- Options
1. $\sqrt{\frac{4L}{g \cos^2 \theta - a_0 \sin \theta \cos \theta}}$
 2. $\sqrt{\frac{2L}{g \sin \theta - a_0 \cos \theta}}$
 3. $\sqrt{\frac{4L}{g \sin 2\theta - a_0 (1 + \cos 2\theta)}}$
 4. $\sqrt{\frac{2L}{g \sin 2\theta - a_0 (1 + \cos 2\theta)}}$

Question Type : MCQ

Question ID : 8606541679

Option 1 ID : 8606545710

Option 2 ID : 8606545711

Option 3 ID : 8606545708

Option 4 ID : 8606545709

Q.45 For a transparent prism, if the angle of minimum deviation is equal to its refracting angle, the refractive index n of the prism satisfies.

Options

1. $1 < n < 2$
2. $\sqrt{2} < n < 2\sqrt{2}$
3. $n \geq 2$
4. $\sqrt{2} < n < 2$

Question Type : MCQ

Question ID : 8606541691

Option 1 ID : 8606545756

Option 2 ID : 8606545758

Option 3 ID : 8606545757

Option 4 ID : 8606545759

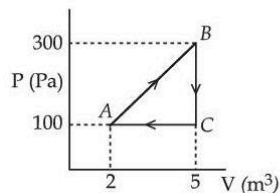
Section : Physics Section B

Q.46 A fly wheel having mass 3 kg and radius 5 m is free to rotate about a horizontal axis. A string having negligible mass is wound around the wheel and the loose end of the string is connected to 3 kg mass. The mass is kept at rest initially and released. Kinetic energy of the wheel when the mass descends by 3 m is _____. J. ($g = 10 \text{ m/s}^2$)

Question Type : SA

Question ID : 8606541696

Q.47 A thermodynamic system is taken through the cyclic process ABC as shown in the figure. The total work done by the system during the cycle ABC is _____. J.



Question Type : SA

Question ID : 8606541699

Q.48 An inductor stores 16 J of magnetic field energy and dissipates 32 W of thermal energy due to its resistance when an a.c. current of 2 A (rms) and frequency 50 Hz flows through it. The ratio of inductive reactance to its resistance is _____. ($\pi = 3.14$)

Question Type : SA

Question ID : 8606541698

Q.49 A beam of light consisting of wavelengths 650 nm and 550 nm illuminates the Young's double slits with separation of 2 mm such that the interference fringes are formed on a screen, placed at a distance of 1.2 m from the slits. The least distance of a point from the central maximum, where the bright fringes due to both the wavelengths coincide, is _____ $\times 10^{-5}$ m.

Question Type : SA

Question ID : 8606541700

Q.50 Two tuning forks *A* and *B* are sounded together giving rise to 8 beats in 2 s. When fork *A* is loaded with wax, the beat frequency is reduced to 4 beats in 2 s. If the original frequency of tuning fork *B* is 380 Hz then original frequency of tuning fork *A* is _____ Hz.

Question Type : **SA**

Question ID : **8606541697**

Section : **Chemistry Section A**

Q.51 The wavelength of photon '*A*' is 400 nm. The frequency of photon '*B*' is 10^{16} s^{-1} . The wave number of photon '*C*' is 10^4 cm^{-1} . The correct order of energy of these photons is :

Options

1. $A > B > C$
2. $C > B > A$
3. $A > C > B$
4. $B > A > C$

Question Type : **MCQ**

Question ID : **8606541702**

Option 1 ID : **8606545785**

Option 2 ID : **8606545786**

Option 3 ID : **8606545788**

Option 4 ID : **8606545787**

Q.52 Consider the elements N, P, O, S, Cl and F. The number of valence electrons present in the elements with most and least metallic character from the above list is respectively.

Options

1. 5 and 6
2. 5 and 7
3. 7 and 5
4. 6 and 7

Question Type : **MCQ**

Question ID : **8606541707**

Option 1 ID : **8606545808**

Option 2 ID : **8606545807**

Option 3 ID : **8606545805**

Option 4 ID : **8606545806**

Q.53 Given below are two statements :

Statement I : The increasing order of boiling point of hydrogen halides is $\text{HCl} < \text{HBr} < \text{HI} < \text{HF}$.

Statement II : The increasing order of melting point of hydrogen halides is $\text{HCl} < \text{HBr} < \text{HF} < \text{HI}$.

In the light of the above statements, choose the **correct** answer from the options given below :

Options

1. **Statement I** is false but **Statement II** is true
2. Both **Statement I** and **Statement II** are true
3. **Statement I** is true but **Statement II** is false
4. Both **Statement I** and **Statement II** are false

Question Type : **MCQ**

Question ID : **8606541708**

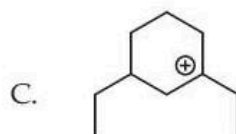
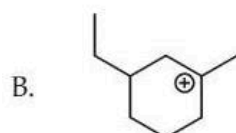
Option 1 ID : **8606545812**

Option 2 ID : **8606545809**

Option 3 ID : **8606545811**

Option 4 ID : **8606545810**

Q.54 The cyclic cations having the same number of hyperconjugation are :



Choose the **correct** answer from the options given below :

Options

1. A and B Only
2. A and C Only
3. B and C Only
4. A, C and D only

Question Type : **MCQ**

Question ID : **8606541712**

Option 1 ID : **8606545825**

Option 2 ID : **8606545828**

Option 3 ID : **8606545827**

Option 4 ID : **8606545826**

Q.55 A student performed analysis of aliphatic organic compound 'X' which on analysis gave C = 61.01%, H = 15.25%, N = 23.74%.

This compound, on treatment with $\text{HNO}_2/\text{H}_2\text{O}$ produced another compound 'Y' which did not contain any nitrogen atom. However, the compound 'Y' upon controlled oxidation produced another compound 'Z' that responded to iodoform test.

The structure of 'X' is :

Options

1. $\begin{array}{c} \text{CH}_3 \backslash \\ \text{CH} - \text{NH}_2 \\ \text{CH}_3 / \end{array}$
2. $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$
3. $\begin{array}{c} \text{CH}_3 - \text{CH}_2\text{CH} - \text{CH}_3 \\ | \\ \text{NH}_2 \end{array}$
4. $\begin{array}{c} \text{Ph} - \text{CH} - \text{NH}_2 \\ | \\ \text{CH}_3 \end{array}$

Question Type : MCQ

Question ID : 8606541717

Option 1 ID : 8606545848

Option 2 ID : 8606545845

Option 3 ID : 8606545847

Option 4 ID : 8606545846

Q.56 Identify the **correct** statements :

The presence of $-\text{NO}_2$ group in benzene ring

- A. activates the ring towards electrophilic substitutions.
- B. deactivates the ring towards electrophilic substitutions.
- C. activates the ring towards nucleophilic substitutions.
- D. deactivates the ring towards nucleophilic substitutions.

Choose the **correct** answer from the options given below :

Options

1. B and C Only
2. B and D Only
3. A and D Only
4. C and A Only

Question Type : MCQ

Question ID : 8606541713

Option 1 ID : 8606545831

Option 2 ID : 8606545832

Option 3 ID : 8606545829

Option 4 ID : 8606545830

Q.57 A student has been given 0.314 g of an organic compound and asked to estimate Sulphur. During the experiment, the student has obtained 0.4813 g of barium sulphate. The percentage of sulphur present in the compound is _____. (Given Molar mass in g mol^{-1} S : 32, BaSO_4 : 233)

Options

1. 42.10%
2. 48.24%
3. 21.05%
4. 63.15%

Question Type : MCQ

Question ID : 8606541720

Option 1 ID : 8606545857

Option 2 ID : 8606545859

Option 3 ID : 8606545860

Option 4 ID : 8606545858

Q.58

Match List - I with List - II according to shape.

List - I

- A. XeO_3
B. XeF_2
C. XeO_2F_2
D. XeOF_4

List - II

- I. BrF_5
II. NH_3
III. $[\text{I}_3]^-$
IV. SF_4

Choose the **correct** answer from the options given below :

Options

1. A-II, B-III, C-IV, D-I
2. A-II, B-III, C-I, D-IV
3. A-III, B-II, C-IV, D-I
4. A-II, B-I, C-III, D-IV

Question Type : MCQ

Question ID : 8606541706

Option 1 ID : 8606545804

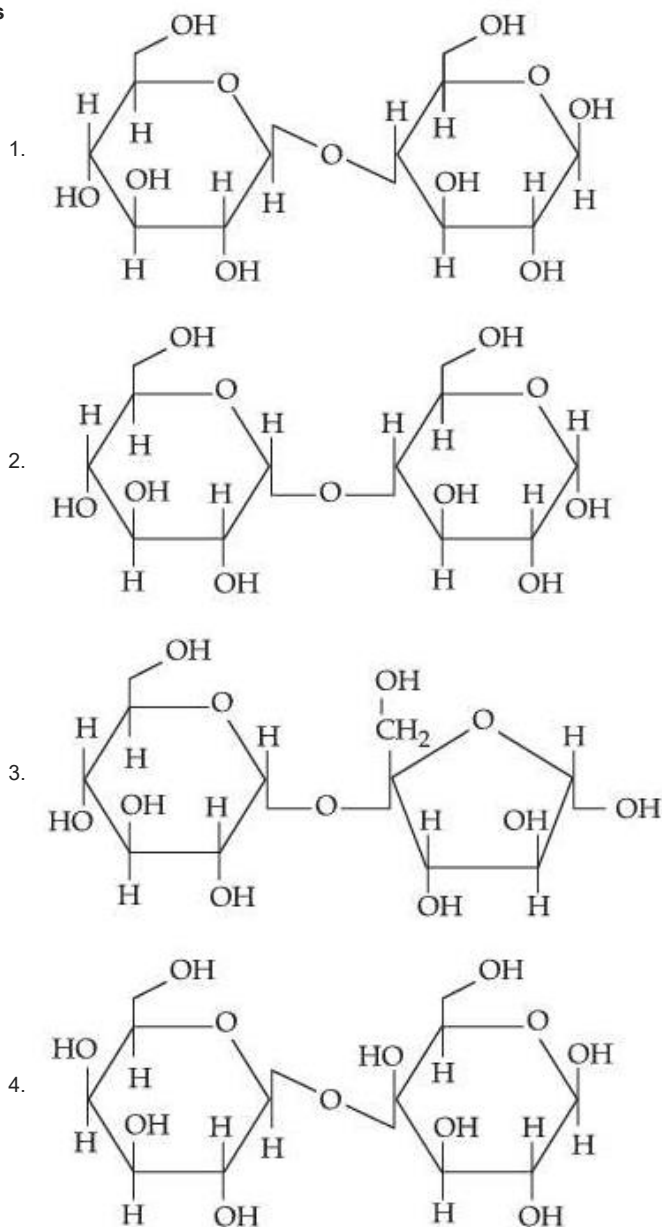
Option 2 ID : 8606545802

Option 3 ID : 8606545803

Option 4 ID : 8606545801

Q.59 Structures of four disaccharides are given below. Among the given disaccharides, the non-reducing sugar is :

Options



Question Type : **MCQ**

Question ID : **8606541719**

Option 1 ID : **8606545856**

Option 2 ID : **8606545853**

Option 3 ID : **8606545854**

Option 4 ID : **8606545855**

Q.60 Total number of alkali insoluble solid sulphonamides obtained by reaction of given amines with Hinsberg's reagent is _____.

Aniline, N-Methylaniline, Methanamine, N, N – Dimethylmethanamine, N-Methyl methanamine, Phenylmethanamine, N-propylaniline, N-phenylaniline, N, N-Dimethylaniline, Allyl amine, Isopropyl amine

Options

1. 4
2. 8
3. 2
4. 5

Question Type : MCQ

Question ID : 8606541718

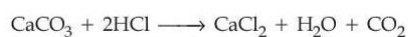
Option 1 ID : 8606545850

Option 2 ID : 8606545851

Option 3 ID : 8606545852

Option 4 ID : 8606545849

Q.61 For the given reaction;



If 90 g CaCO_3 is added to 300 mL of HCl which contains 38.55% HCl by mass and has density 1.13 g mL^{-1} , then which of the following option is correct ?

Given molar mass of H, Cl, Ca and O are 1, 35.5, 40 and 16 g mol^{-1} respectively.

Options

1. 97.30 g of HCl reacted
2. 60.32 g of HCl remains unreacted
3. 32.85 g of CaCO_3 remains unreacted
4. 64.97 g of HCl remains unreacted

Question Type : MCQ

Question ID : 8606541701

Option 1 ID : 8606545784

Option 2 ID : 8606545781

Option 3 ID : 8606545783

Option 4 ID : 8606545782

Q.62 Consider the following aqueous solutions.

- I. 2.2 g Glucose in 125 mL of solution.
- II. 1.9 g Calcium chloride in 250 mL of solution.
- III. 9.0 g Urea in 500 mL of solution.
- IV. 20.5 g Aluminium sulphate in 750 mL of solution.

The correct increasing order of boiling point of these solutions will be :

[Given : Molar mass in g mol^{-1} : H = 1, C = 12, N = 14, O = 16, Cl = 35.5, Ca = 40, Al = 27 and S = 32]

Options

- 1. $\text{II} < \text{III} < \text{I} < \text{IV}$
- 2. $\text{II} < \text{III} < \text{IV} < \text{I}$
- 3. $\text{I} < \text{II} < \text{III} < \text{IV}$
- 4. $\text{III} < \text{I} < \text{II} < \text{IV}$

Question Type : **MCQ**

Question ID : **8606541704**

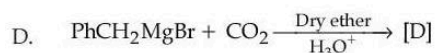
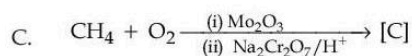
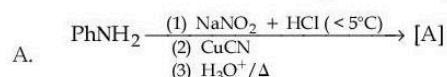
Option 1 ID : **8606545795**

Option 2 ID : **8606545796**

Option 3 ID : **8606545793**

Option 4 ID : **8606545794**

Q.63 The correct order of acidic strength of the major products formed in the given reactions, is :



Choose the correct answer from the options given below :

Options

- 1. $\text{C} > \text{A} > \text{D} > \text{B}$
- 2. $\text{A} > \text{D} > \text{C} > \text{B}$
- 3. $\text{C} > \text{B} > \text{A} > \text{D}$
- 4. $\text{A} > \text{D} > \text{B} > \text{C}$

Question Type : **MCQ**

Question ID : **8606541716**

Option 1 ID : **8606545843**

Option 2 ID : **8606545842**

Option 3 ID : **8606545844**

Option 4 ID : **8606545841**

Q.64

The plot of $\log_{10} K$ vs $\frac{1}{T}$ gives a straight line. The intercept and slope respectively are
(where K is equilibrium constant).

Options

1. $-\frac{\Delta H^0}{2.303R}$, $\frac{\Delta S^0}{2.303R}$
2. $\frac{\Delta S^0}{2.303R}$, $-\frac{\Delta H^0}{2.303R}$
3. $\frac{2.303R}{\Delta H^0}$, $\frac{2.303R}{\Delta S^0}$
4. $-\frac{\Delta S^0 R}{2.303}$, $\frac{\Delta H^0 R}{2.303}$

Question Type : MCQ

Question ID : 8606541703

Option 1 ID : 8606545789

Option 2 ID : 8606545792

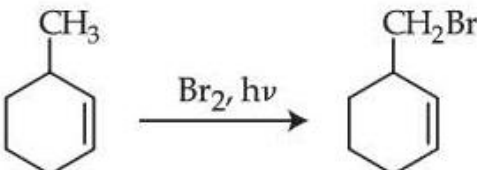
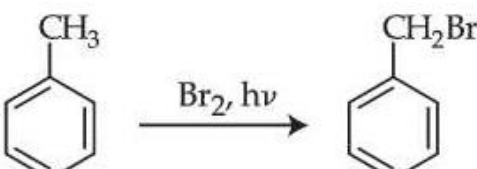
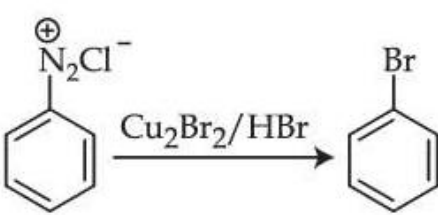
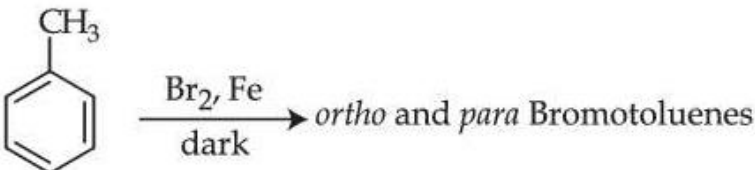
Option 3 ID : 8606545791

Option 4 ID : 8606545790

Q.65

Which of the following reaction is NOT correctly represented ?

Options

1. 
2. 
3. 
4. 

Question Type : MCQ

Question ID : 8606541715

Option 1 ID : 8606545837

Option 2 ID : 8606545838

Option 3 ID : 8606545840

Option 4 ID : 8606545839

Q.66 The correct increasing order of spin-only magnetic moment values of the complex ions $[\text{MnBr}_4]^{2-}$ (A), $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ (B), $[\text{Ni}(\text{CN})_4]^{2-}$ (C) and $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$ (D) is :

Options

1. $C < B < D < A$
2. $C = D < B < A$
3. $A = B < C < D$
4. $A = B < D < C$

Question Type : MCQ

Question ID : 8606541710

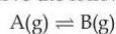
Option 1 ID : 8606545820

Option 2 ID : 8606545819

Option 3 ID : 8606545817

Option 4 ID : 8606545818

Q.67 Observe the following equilibrium in a 1 L flask.



At T(K), the equilibrium concentrations of A and B are 0.5 M and 0.375 M respectively. 0.1 moles of A is added into the flask and heated to T(K) to establish the equilibrium again. The new equilibrium concentrations (in M) of A and B are respectively

Options

1. 0.742, 0.557.
2. 0.367, 0.275.
3. 0.53, 0.4.
4. 0.557, 0.418.

Question Type : MCQ

Question ID : 8606541705

Option 1 ID : 8606545798

Option 2 ID : 8606545799

Option 3 ID : 8606545800

Option 4 ID : 8606545797

Q.68 Consider the following statements about manganate and permanganate ions. Identify the **correct** statements.

- A. The geometry of both manganate and permanganate ions is tetrahedral.
- B. The oxidation states of Mn in manganate and permanganate are +7 and +6, respectively.
- C. Oxidation of Mn(II) salt by peroxodisulphate gives manganate ion as the final product.
- D. Manganate ion is paramagnetic and permanganate ions is diamagnetic.
- E. Acidified permanganate ion reduces oxalate, nitrite and iodide ions.

Choose the **correct** answer from the options given below :

Options

- 1. A, C and D Only
- 2. A and D Only
- 3. A, B and C Only
- 4. A, D and E Only

Question Type : **MCQ**

Question ID : **8606541709**

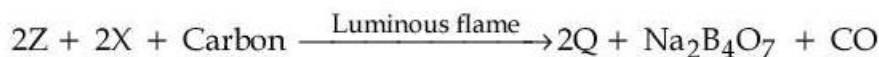
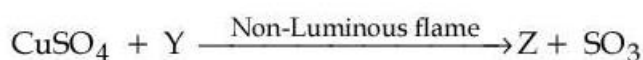
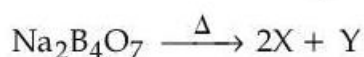
Option 1 ID : **8606545815**

Option 2 ID : **8606545816**

Option 3 ID : **8606545813**

Option 4 ID : **8606545814**

Q.69 Consider the following reactions



The oxidation states of Cu in Z and Q, respectively are :

Options

- 1. +1 and +1
- 2. +2 and +2
- 3. +1 and +2
- 4. +2 and +1

Question Type : **MCQ**

Question ID : **8606541711**

Option 1 ID : **8606545823**

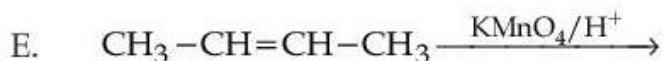
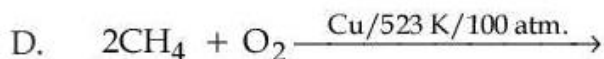
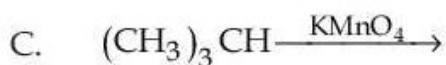
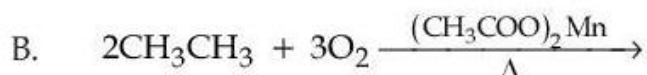
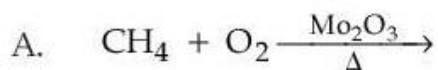
Option 2 ID : **8606545821**

Option 3 ID : **8606545824**

Option 4 ID : **8606545822**

Q.70

The reactions which produce alcohol as the product are :



Choose the **correct** answer from the options given below :

Options

1. C and D Only
2. B, D and E Only
3. A, C and E Only
4. A and D Only

Question Type : MCQ

Question ID : 8606541714

Option 1 ID : 8606545835

Option 2 ID : 8606545834

Option 3 ID : 8606545836

Option 4 ID : 8606545833

Section : Chemistry Section B

Q.71

A $\longrightarrow$ B (first reaction)

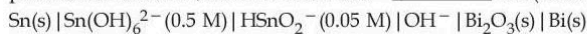
C $\longrightarrow$ D (second reaction)

Consider the above two first-order reactions. The rate constant for first reaction at 500 K is double of the same at 300 K. At 500 K, 50% of the reaction becomes complete in 2 hour. The activation energy of the second reaction is half of that of first reaction. If the rate constant at 500 K of the second reaction becomes double of the rate constant of first reaction at the same temperature; then rate constant for the second reaction at 300 K is $\times 10^{-1}\text{hour}^{-1}$ (nearest integer).

Question Type : SA

Question ID : 8606541724

- Q.72** A volume of x mL of 5 M NaHCO_3 solution was mixed with 10 mL of 2 M H_2CO_3 solution to make an electrolytic buffer. If the same buffer was used in the following electrochemical cell to record a cell potential of 235.3 mV, then the value of $x =$ _____ mL (nearest integer).



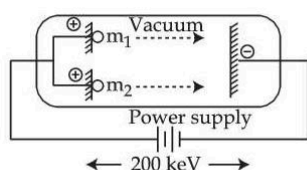
Consider upto one place of decimal for intermediate calculations

$$\left[\begin{array}{l} \text{Given :} \quad E^\circ_{\text{HSnO}_2^-/\text{Sn(OH)}_6^{2-}} = -0.9 \text{ V} \\ E^\circ_{\text{Bi}_2\text{O}_3/\text{Bi}} = -0.44 \text{ V} \\ \text{pK}_a(\text{H}_2\text{CO}_3) = 6.11 \\ \frac{2.303 RT}{F} = 0.059 \text{ V} \\ \text{Antilog}(1.29) = 19.5 \end{array} \right]$$

Question Type : SA

Question ID : 8606541723

- Q.73** Two positively charged particles m_1 and m_2 have been accelerated across the same potential difference of 200 keV as shown below.



[Given mass of $m_1 = 1$ amu and $m_2 = 4$ amu]

The deBroglie wavelength of m_1 will be x times of m_2 . The value of x is _____ (nearest integer)

Question Type : SA

Question ID : 8606541721

- Q.74** For strong electrolyte Λ_m increases slowly with dilution and can be represented by the equation

$$\Lambda_m = \Lambda_m^\circ - A c^{1/2}$$

Molar conductivity values of the solutions of strong electrolyte AB at 18°C are given below :

c [mol L ⁻¹]	0.04	0.09	0.16	0.25
Λ_m [S cm ² mol ⁻¹]	96.1	95.7	95.3	94.9

The value of constant A based on the above data [in S cm² mol⁻¹/(mol/L)^{1/2}] unit is _____.

Question Type : SA

Question ID : 8606541722

- Q.75** The number of isoelectronic species among Sc^{3+} , Cr^{2+} , Mn^{3+} , Co^{3+} and Fe^{3+} is 'n'. If 'n' moles of AgCl is formed during the reaction of complex with formula $\text{CoCl}_3(\text{en})_2\text{NH}_3$ with excess of AgNO_3 solution, then the number of electrons present in the t_{2g} orbital of the complex is _____.

Question Type : SA

Question ID : 8606541725